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Theme 1.2 Market Failure — Edexcel B

1. What Is Market Failure?

Market failure: When the free market mechanism leads to a misallocation of resources — the market either over-produces or under-produces certain goods relative to the socially optimal level. Resources are not allocated efficiently, meaning social welfare is not maximised.

1.1 Types of Market Failure

- **Externalities** — costs or benefits that affect third parties not involved in the transaction.
- **Public goods** — goods that the market fails to provide due to non-excludability and non-rivalry.
- **Merit and demerit goods** — goods that are under-consumed or over-consumed due to information failures and externalities.
- **Information failure** — when economic agents lack complete or accurate information.
- **Monopoly power** — when firms restrict output and raise prices above competitive levels.
- **Factor immobility** — when factors of production cannot easily move between uses.
- **Inequality** — the free market may produce an inequitable distribution of income and wealth.

Key terminology:
Marginal Private Cost (MPC): The cost to the producer of producing one additional unit.
Marginal Private Benefit (MPB): The benefit to the consumer of consuming one additional unit.
Marginal External Cost (MEC): The cost imposed on third parties from one additional unit.
Marginal External Benefit (MEB): The benefit to third parties from one additional unit.
Marginal Social Cost (MSC): MPC + MEC — the full cost to society.
Marginal Social Benefit (MSB): MPB + MEB — the full benefit to society.

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$$\text{MSC} = \text{MPC} + \text{MEC} \qquad \text{MSB} = \text{MPB} + \text{MEB}$$

The socially optimal output is where **MSC = MSB**. If the market produces at a different level, there is a welfare loss (deadweight loss).

2. Externalities

Externality: A cost or benefit that arises from an economic transaction and falls on a third party who is not directly involved in the transaction. Externalities cause a divergence between private and social costs or benefits, leading to market failure.

2.1 Negative Externalities of Production

When the production of a good imposes costs on third parties. The MPC understates the true cost to society — the MSC is greater than the MPC. The market over-produces relative to the social optimum.

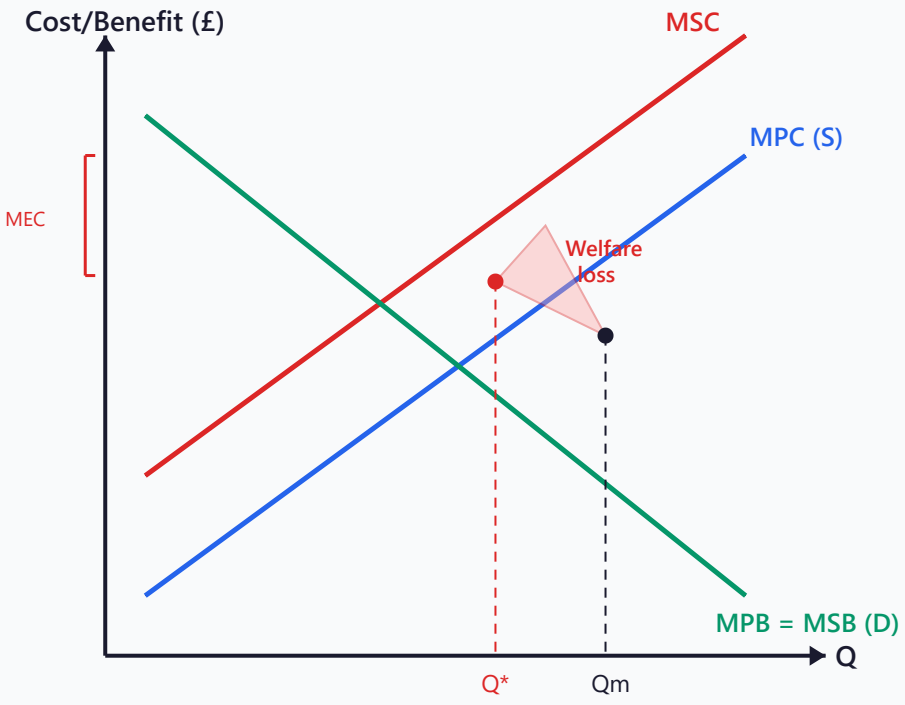

Figure 1: Negative externality of production — MSC > MPC. Market over-produces at Qm; socially optimal output is Q. The shaded triangle is the welfare (deadweight) loss.*

Analysis: The vertical distance between MSC and MPC represents the marginal external cost (MEC) — the cost imposed on third parties per unit. The market produces at Qm (where MPC = MPB) but the social optimum is Q* (where MSC = MSB). Between Q* and Qm, the social cost of each unit exceeds its social benefit, creating a welfare loss.

Case Study: Air Pollution from UK Industry

The UK's manufacturing and energy sectors impose significant negative externalities through air pollution. According to Public Health England, air pollution contributes to approximately 28,000-36,000 deaths per year in the UK. The market price of goods produced by polluting industries does not reflect the health costs borne by local communities, NHS treatment costs, or environmental damage. This is why the UK participates in the Emissions Trading Scheme — to internalise these external costs and move production closer to the socially optimal level.

2.2 Negative Externalities of Consumption

When consumption of a good imposes costs on third parties. The MPB overstates the true benefit to society — the MSB is less than the MPB. The market over-consumes relative to the social optimum.

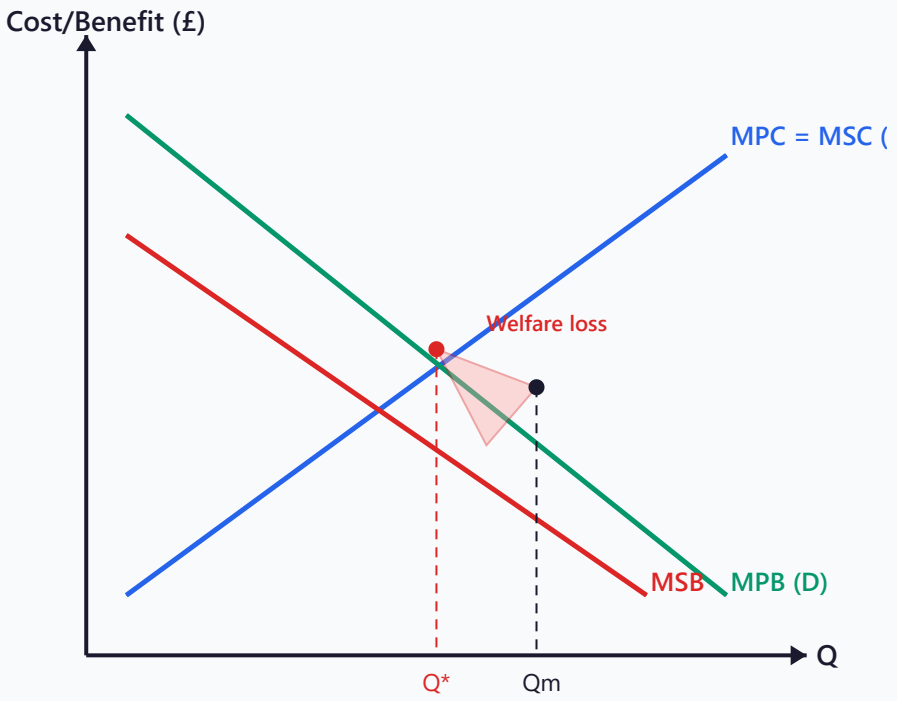

Figure 2: Negative externality of consumption — MSB < MPB. Market over-consumes at Qm; social optimum is Q.*

Examples: Smoking (passive smoking harms others, NHS costs), alcohol consumption (antisocial behaviour, policing costs, NHS costs), driving (congestion, pollution, accidents affecting third parties).

2.3 Positive Externalities of Production

When production of a good creates benefits for third parties. The MPC overstates the true cost to society — the MSC is less than the MPC. The market under-produces relative to the social optimum.

Example: A firm that trains its workers generates positive externalities — those workers may move to other firms, spreading skills across the economy. Bee-keepers produce honey, but their bees also pollinate neighbouring crops, benefiting farmers.

2.4 Positive Externalities of Consumption

When consumption of a good creates benefits for third parties. The MPB understates the true benefit to society — the MSB is greater than the MPB. The market under-consumes relative to the social optimum.

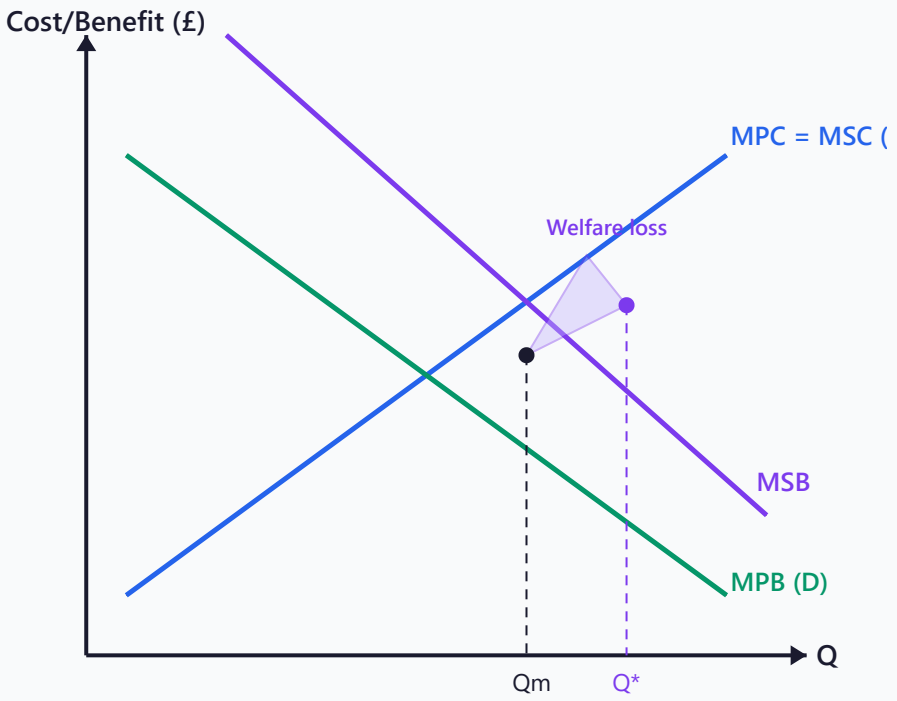

Figure 3: Positive externality of consumption — MSB > MPB. Market under-consumes at Qm; social optimum is Q.*

Examples: Education (higher productivity benefits employers and economy), vaccination (herd immunity protects others), healthcare (healthier workers are more productive).

Case Study: COVID-19 Vaccination Programme

The UK's COVID-19 vaccination programme is a textbook example of positive consumption externalities. Each person vaccinated not only protects themselves but also reduces the likelihood of transmitting the virus to others (herd immunity). If left to the free market, the vaccine would have been under-consumed — many people would weigh only their private costs and benefits, ignoring the social benefits. The government intervened by making the vaccine free at the point of use and running mass vaccination centres, moving consumption towards the socially optimal level.

2.5 Summary of Externalities

Type	Key Relationship	Market Outcome	Solution Direction

Negative production externality	MSC > MPC	Over-production	Tax, regulation, tradeable permits
Negative consumption externality	MSB < MPB	Over-consumption	Tax, ban, education, regulation
Positive production externality	MSC < MPC	Under-production	Subsidy to producers
Positive consumption externality	MSB > MPB	Under-consumption	Subsidy, free provision, legislation

3. Public Goods

Public good: A good that has two characteristics — **non-excludability** (it is impossible to prevent anyone from consuming it) and **non-rivalry** (one person's consumption does not reduce the amount available for others). Because of these properties, the free market fails to provide public goods — this is known as the **missing market**.

3.1 The Free-Rider Problem

Because public goods are non-excludable, rational consumers have an incentive to free-ride — to benefit from the good without paying for it. If everyone free-rides, no one pays, and the good is not provided. This is why the free market under-provides or completely fails to provide public goods.

3.2 Examples

Public Goods	Quasi-Public Goods	Private Goods
National defence	Roads (excludable via tolls; rivalrous in congestion)	Food, clothing
Street lighting	Parks (can charge entry; crowding)	Cars, phones
Lighthouses	BBC (could be encrypted; non-rival)	Haircuts

Flood defences	Fire services	Cinema tickets

Quasi-public goods possess characteristics of both public and private goods. They may be partially excludable or partially rivalrous. The government often provides or subsidises them because the market alone would under-provide.

Evaluation: Public Goods

- Technology can change the nature of goods — e.g., encryption makes broadcasting excludable; toll technology makes roads excludable.
- Some goods considered public can be privately provided if clever mechanisms are found (e.g., lighthouses were historically privately funded through port dues).
- Government provision of public goods requires taxation — creating its own distortions and deadweight losses.
- Government may over-provide or mis-allocate resources to public goods (government failure).

4. Merit and Demerit Goods

4.1 Merit Goods

Merit good: A good that is under-consumed in the free market because consumers do not fully appreciate its private benefits (imperfect information) and/or because it generates positive externalities. Society judges that consumption should be higher than the free market level.

Merit goods have two key features causing under-consumption:

1. **Information failure:** Consumers underestimate the private benefits (e.g., a teenager may not appreciate the long-term benefits of education).
2. **Positive externalities:** Consumption generates benefits for third parties that consumers do not consider (e.g., education benefits employers and the economy).

Examples: Education, healthcare, vaccinations, museums, exercise, healthy eating.

4.2 Demerit Goods

Demerit good: A good that is over-consumed in the free market because consumers do not fully appreciate its private costs (imperfect information) and/or because it generates negative externalities.

Demerit goods have two key features causing over-consumption:

1. **Information failure:** Consumers underestimate the private costs (e.g., a smoker may not fully appreciate health risks).
2. **Negative externalities:** Consumption imposes costs on third parties (e.g., passive smoking, drink-driving).

Examples: Tobacco, alcohol, recreational drugs, gambling, junk food.

Exam Tip: The distinction between a negative externality and a demerit good is important. A negative externality is about third-party costs. A demerit good also involves information failure — the consumer themselves does not fully understand the harm. Examiners want you to identify both elements for merit/demerit goods.

Case Study: NHS and Education as Merit Goods

The UK provides the NHS free at the point of use and compulsory education until age 18. Without these interventions, both would be under-consumed — many people would not purchase health insurance or invest in education due to short-termism, information failure (young people underestimating the returns to education), and inability to pay. The NHS costs over 180 billion per year. By making these services free and compulsory, the government moves consumption closer to the socially optimal level.

5. Information Failure

Information failure (asymmetric information): When economic agents lack complete, accurate, or symmetric information to make rational decisions. This leads to a misallocation of resources.

5.1 Types of Information Failure

- **Imperfect information:** Consumers or producers lack knowledge needed for rational decisions (e.g., not knowing the true sugar content of food).
- **Asymmetric information:** One party in a transaction has more information than the other (e.g., a used car seller knows more about the car than the buyer — Akerlof's "market for lemons").
- **Bounded rationality:** Consumers may have information but lack the ability to process it rationally (behavioural economics concept).

5.2 Consequences

- **Adverse selection:** The informed party exploits their advantage before a transaction (e.g., unhealthy people are more likely to buy insurance).
- **Moral hazard:** The informed party changes behaviour after a transaction (e.g., insured drivers take more risks).
- **Market breakdown:** If buyers cannot distinguish quality, good products may be driven out (Gresham's Law applied to markets).

Case Study: Payday Lending and Information Failure

The payday lending market in the UK illustrated severe information failure. Firms like Wonga charged APRs exceeding 4,000%, but consumers often focused only on the short-term cash amount rather than the true cost of borrowing. Many borrowers underestimated the probability of being unable to repay on time, leading to debt spirals. The FCA intervened in 2015 by capping the total cost of a loan at 100% of the amount borrowed and requiring clearer information disclosure. This is a case where imperfect consumer information led to over-consumption of a harmful financial product.

6. Government Intervention to Correct Market Failure

6.1 Indirect Taxes

Indirect tax: A tax on spending — levied on producers but partially passed on to consumers through higher prices. Used to internalise negative externalities by raising the private cost of

production/consumption.

Specific tax: A fixed amount per unit (e.g., 3.50 per bottle of spirits). Causes a parallel upward shift of the supply curve.

Ad valorem tax: A percentage of the price (e.g., 20% VAT). Causes a pivoting shift of the supply curve (larger absolute gap at higher prices).

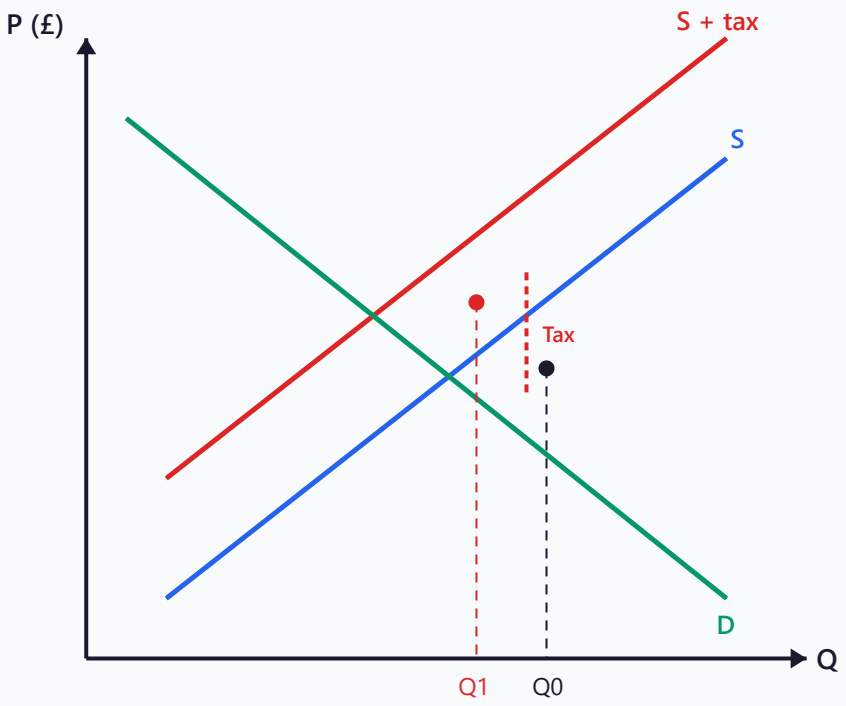

Figure 4: An indirect (specific) tax shifts supply left, raising price and reducing quantity towards the social optimum.

Evaluation: Indirect Taxes

- Effectiveness depends on PED — if demand is inelastic (e.g., cigarettes), tax has little effect on consumption.
- Difficult to calculate the correct tax rate to match the external cost precisely.
- Regressive — takes a larger proportion of income from the poor.
- May encourage black markets and smuggling (e.g., tobacco smuggling from Eastern Europe).
- Generates government revenue that can fund public services.

- May reduce international competitiveness if competitors face lower taxes.

6.2 Subsidies

Subsidy: A payment by the government to producers that reduces the cost of production and shifts the supply curve to the right. Used to encourage production and consumption of goods with positive externalities.

A subsidy effectively lowers the price for consumers and increases output, moving closer to the socially optimal level for goods that are under-consumed or under-produced.

Evaluation: Subsidies

- Opportunity cost — money spent on subsidies could be used elsewhere.
- Difficult to determine the correct subsidy level to match the external benefit.
- May lead to dependency — firms may become reliant on subsidies rather than becoming efficient.
- May create market distortions and unintended consequences.
- Benefits may be captured by producers (higher profits) rather than passed to consumers.

6.3 Maximum and Minimum Prices

Maximum Price (Price Ceiling)

A legally imposed maximum price set below the equilibrium. Purpose: to make goods affordable for consumers (e.g., rent controls).

- Creates excess demand (shortage) — Qd > Qs at the controlled price.
- May lead to black markets, queuing, rationing, reduced quality.
- Producers lose revenue — may exit the market, worsening shortages long-term.

Minimum Price (Price Floor)

A legally imposed minimum price set above equilibrium. Purpose: to protect producers' incomes or discourage consumption (e.g., minimum wage, minimum alcohol pricing).

- Creates excess supply — Qs > Qd at the controlled price.
- Government may need to buy up the surplus (costly).
- Consumers pay more; may be regressive.

6.4 Tradeable Pollution Permits

Tradeable permits: A government sets a total cap on pollution, then issues permits that firms can buy, sell, and trade. Firms that can reduce pollution cheaply sell permits; those that find it expensive to reduce pollution buy permits. This creates a market-based incentive to reduce pollution.

The government gradually reduces the total number of permits, lowering total pollution over time.

Evaluation: Tradeable Permits

- Market-based mechanism — efficient because firms that can reduce pollution cheaply do so first.
- Cap guarantees the total level of pollution reduction (unlike a tax, where the quantity effect is uncertain).
- Price volatility in the permit market can make business planning difficult.
- Difficult to set the correct cap — too many permits and pollution remains too high.
- Only works if monitored and enforced effectively.
- May just shift pollution to countries without such schemes (carbon leakage).

6.5 Regulation and Legislation

The government can directly ban or restrict activities that cause market failure:

- Banning smoking in public places (Health Act 2006).
- Compulsory seatbelt and crash-helmet laws.
- Environmental regulations (emissions standards, waste disposal rules).

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- Competition law (preventing monopoly abuse).
- Compulsory education until age 18.

6.6 Provision of Information

The government can address information failures by requiring or funding information provision:

- Health warnings on cigarette packets and alcohol.
- Nutritional labelling on food.
- Public health campaigns (e.g., Change4Life, 5-a-day).
- Energy efficiency ratings on appliances.
- Financial literacy programmes.

6.7 State Provision

The government directly provides goods and services when the market would under-provide or fail entirely:

- Public goods (defence, street lighting, flood defences).
- Merit goods (NHS, state education).
- Natural monopolies (historically rail, water, energy).

6.8 Property Rights and the Coase Theorem

If property rights are clearly defined and transaction costs are low, private bargaining between parties can solve externality problems without government intervention. For example, if residents downstream have a legal right to clean water, they can negotiate compensation from a polluting factory — internalising the externality.

However, in practice this often fails because transaction costs are high, property rights are unclear, and large numbers of affected parties make negotiation impractical.

7. Government Failure

Government failure: When government intervention to correct market failure leads to a worse allocation of resources than the free market would have achieved — creating a net welfare loss

rather than a gain.

7.1 Causes of Government Failure

- **Unintended consequences:** Policies may have effects that were not anticipated. For example, rent controls designed to help tenants may reduce the supply of rental housing, making the problem worse long-term.
- **Distortion of price signals:** Intervention (taxes, subsidies, price controls) distorts the information that prices convey, leading to misallocation.
- **Excessive administrative costs:** The costs of implementing, monitoring, and enforcing policies may outweigh the benefits.
- **Information gaps:** The government may lack the information needed to set the right tax rate, subsidy level, or regulation.
- **Political self-interest:** Politicians may pursue policies that win votes rather than maximise social welfare (public choice theory). Short election cycles encourage short-termism.
- **Regulatory capture:** Regulators may become too close to the industries they regulate, serving industry interests rather than the public interest.
- **Moral hazard:** Government safety nets (bailouts, welfare) may reduce incentives for responsible behaviour.
- **Disincentive effects:** High taxes may discourage work, enterprise, and investment — reducing economic growth.

Case Study: EU Common Agricultural Policy (CAP)

The CAP is a frequently cited example of government failure. Minimum prices (price floors) for agricultural products led to massive over-production — the so-called "butter mountains" and "wine lakes." The EU spent billions buying up surplus produce, much of which was destroyed or dumped on world markets, harming farmers in developing countries. The policy was intended to protect European farmers' incomes but led to higher food prices for consumers, environmental damage from intensive farming, and distorted global trade. While the CAP has been reformed significantly, it still absorbs about a third of the EU budget.

Evaluation: Market Failure vs Government Failure

- The existence of market failure does not automatically justify intervention — the cure must not be worse than the disease.
- A cost-benefit analysis should compare the welfare loss from market failure with the potential welfare loss from government failure.
- Some economists (e.g., Austrian school) argue that markets self-correct over time and government intervention is almost always harmful.
- Others (e.g., Keynesian) argue that market failures are pervasive and government has an essential corrective role.
- In practice, the best approach is often a combination of market mechanisms and targeted intervention.

Exam Tip: When evaluating any government intervention, always discuss the potential for government failure. Consider: (1) Is the information available to set the right policy level? (2) What are the unintended consequences? (3) What are the administrative costs? (4) Are there political incentives that distort the policy? This balanced analysis is essential for top marks on evaluation questions.

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